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ASSESSMENT OF SYNERGISTIC INTERACTION BETWEEN ALLIUM SATIVUM EXTRACT, HONEY, AND AMOXICILLIN AGAINST MULTI-DRUG RESISTANT (MDR) CLINICAL ISOLATES OF ESCHERICHIA COLI

Anirban Maity - Research Scholar

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Sandeep Kumar Mishra - Assistant Professor

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

Antimicrobial resistance (AMR) is one of the key global public health challenges that diminishes the effectiveness of traditional antibiotics (WHO, 2015; Ventola, 2015; Prestinaci et al., 2015). The Gram-negative bacterium Escherichia coli is a common organism linked to the development of MDR, causing therapeutic failure (Kaper et al., 2004; Tadesse et al., 2012). In this experiment, the antimicrobial effect and synergy between Allium sativum extract, honey, and amoxicillin against MDR strains of E. coli were tested. The antibiotic sensitivity test was conducted using the Kirby–Bauer disk diffusion technique, while the MIC was obtained via the broth microdilution method (CLSI, 2020; Andrews, 2001). The checkerboard approach was used to determine the synergistic interaction between the two drugs based on the FICI (Odds, 2003; Eliopoulos & Moellering, 2005). It was found that there was 70% resistance to amoxicillin. Synergy was observed through the decreased MIC by 32 times.

Keywords: *Antimicrobial resistance (AMR), Multidrug resistant E. coli, garlic, honey, amoxicillin, synergy, Minimum Inhibitory Concentration (MIC), Fractional Inhibitory Concentration Index (FICI).*

Introduction

One of the most concerning issues of modern times is antimicrobial resistance, which has greatly compromised the action of conventional antibiotics against numerous infectious diseases. AMR contributes substantially to morbidity, mortality, and higher healthcare costs throughout the world (Ventola, 2015; Prestinaci et al., 2015; World Health Organization, 2015). The rapid appearance and spread of resistant bacteria have been caused by their overuse and misuse, which creates a selection pressure favoring resistant strains (Holmes et al., 2016; Davies & Davies, 2010). Thus, infections that can be easily treated now require special measures and novel methods.

Escherichia coli belongs to the group of pathogens that causes numerous infections, including UTI, neonatal meningitis, septicemia, and gastroenterological conditions (Kaper et al., 2004; Tadesse et al., 2012). In addition, uropathogenic E. coli is responsible for up to 80–90% cases of community-acquired UTI. Furthermore, the development of MDR strains complicates treatment as these strains have high-level resistance not only to one class of antibiotics but to many, including β-lactams, fluoroquinolones, and aminoglycosides (Poirel et al., 2018; Magiorakos et al., 2012).

The main mechanisms of antimicrobial resistance in E. coli include producing β-lactamase, changing PBP, reduction in permeability due to the loss of porins, and efflux of antibiotics (Munita & Arias, 2016; Blair et al., 2015; Tenover, 2006). However, the production of ESBLs can be considered a primary mechanism because of its impact on the effectiveness of β-lactam antibiotics, including amoxicillin and cephalosporins (Bush & Bradford, 2016; Paterson & Bonomo, 2005). Therefore, the therapeutic effect of the majority of antibiotics has considerably decreased, and alternative treatments must be found.

Amoxicillin is a commonly used β-lactam antibiotic that acts by blocking bacterial cell wall formation via binding to PBPs (Bush & Bradford, 2016). Despite its broad spectrum and widespread application, resistance to amoxicillin has emerged due to the development of β-lactamases and other mechanisms of resistance (Poirel et al., 2018; Munita & Arias, 2016). The increasing prevalence of antibiotic resistance has spurred the need to explore alternative approaches for restoring the efficacy of amoxicillin. Natural product combinations represent one promising area being explored for this purpose.

Natural products have long been applied in traditional medicines and are once again receiving attention for their possible roles in antimicrobial treatment due to their various bioactive compounds and multifaceted

modes of actions (Cowan, 1999). Among the many natural products that have been studied, Allium sativum (garlic) is among the most widely studied and has been reported to have antimicrobial, anti-inflammatory, and antioxidant properties (Rivlin, 2001; Bayan et al., 2014). The main biologically active ingredient of garlic, allicin, is produced enzymatically by crushing garlic bulbs and has demonstrated significant antimicrobial effects against various Gram-positive and Gram-negative bacteria, even those of the multi-drug-resistant type (Borlinghaus et al., 2014; Ankri & Mirelman, 1999). The antimicrobial activities of allicin can be attributed to its interactions with thiol enzymes, destruction of bacterial cell walls, and inhibition of nucleic acid biosynthesis (Feldberg et al., 1988; Tsao & Yin, 2001).

In addition, honey is known for its anti-microbial properties since ancient times and is currently becoming widely researched in regard to its properties as a therapeutic agent (Molan, 1992; Almasaudi, 2021). Antibacterial activity of honey may be explained by various aspects, namely high osmolality, low pH, hydrogen peroxide content, and bioactivity of compounds such as phenolics and methylglyoxal (Cheesman et al., 2017; Kwakman et al., 2010). Most importantly, it has been found that honey is efficient in destroying antibiotic resistant strains as well as biofilms produced by these strains, which leads to chronicity and persistence of infection (Lu et al., 2011; Lee et al., 2011). The complex nature of the mode of action of honey makes it unlikely for bacteria to develop resistance, therefore, this agent is a good candidate for combinations.

A synergistic antimicrobial therapy refers to a strategy where two or more substances work together to achieve greater antibacterial effect than each of the substances alone (Hemaiswarya et al., 2008; Eliopoulos & Moellering, 2005). Such treatment has several benefits associated with it, including decreased antibiotic dose needed for treatment, decreased toxicity, and a reduced rate of resistance developing. Moreover, natural products have proven to be very successful in enhancing the effects of traditional drugs by increasing their membrane permeability, blocking resistant enzyme systems, or disrupting biofilms (Jenkins & Cooper, 2012; Müller et al., 2013).

Although there has been much research done proving antibacterial properties of both garlic and honey separately, there has not been much information about a potential synergistic effect of this couple of agents along with a common antibiotic used to treat MDR infections.

Therefore, the purpose of this research project is to examine the antibacterial activity as well as a potential synergistic interaction of garlic, honey, and amoxicillin against MDR Escherichia coli strains. Specifically,

MICs will be determined, synergistic interaction between the substances evaluated with a checkerboard assay, and synergy quantified with FICI.

Materials and Methods

Study Design

The current study was conducted using an in vitro experimental approach to investigate the antibacterial efficacy and interaction between Allium sativum (garlic), honey, and amoxicillin on multiple drug resistance strains of Escherichia coli. The study made use of conventional methods such as antibiotic sensitivity testing, minimum inhibitory concentration (MIC), and checkerboard method for synergy testing, as suggested by the standards provided by Clinical and Laboratory Standards Institute (CLSI, 2020; Andrews, 2001; and Odds, 2003).

Study Location and Duration

The experiments were performed in the Pharmaceutical Microbiology Lab, Faculty of Pharmacy, Kalinga University, Atal Nagar, India, in a controlled laboratory environment. All steps were carried out using aseptic techniques and standard microbiology protocols for consistency and reliability of outcomes (CLSI, 2020).

Bacterial Isolates

The clinical isolates of E. coli were obtained from Neuberg Diagnostics, Sankar Nagar, Raipur, Chhattisgarh, India. These bacteria had been studied before and were known to be resistant to many antibiotics. E. coli is an important pathogen that causes diseases in both hospital and community settings, which makes it a good organism to study AMR (Kaper et al., 2004; Tadesse et al., 2012).

These bacteria were stored in nutrient agar slants, and were revived and cultured before being used in experiments.

Culture Media and Reagents

The following microbiological media and solutions were utilized in the experiment:

- Mueller-Hinton Agar (MHA) for antibiotic sensitivity test
- Mueller-Hinton Broth (MHB) for determining MIC

- Nutrient agar for maintaining cultures

The choice of Mueller-Hinton media was based on its consistency and appropriateness for performing antimicrobial susceptibility test according to CLSI guidelines (CLSI, 2020).

Other chemicals and solutions used were:

- Powder form of Amoxicillin antibiotic
- Garlic (Allium sativum) cloves
- Honey (natural/raw)
- Dimethyl sulfoxide (DMSO)
- Saline solution
- Distilled water
- 0.5 McFarland standard

Preparation of Garlic Extract

Garlic (Allium sativum) cloves were collected from the market named Palaud, Naya Paipur, Chhattisgarh, India, and authenticated from the Botany department of Kalinga University. Garlic cloves were washed, peeled, and accurately weighted. About 100 g of garlic cloves was ground in a sterile mortar with a pestle along with 95% ethyl alcohol. The mixture was left undisturbed for 72 hours during which the sample was intermittently shaken for extracting biologically active compounds.

Garlic extract was filtered through muslin cloth and then through Whatman No.1 filter paper. Garlic extract was concentrated in a water bath at 40°C and kept at 4°C till usage (Ankri & Mirelman, 1999; Borlinghaus et al., 2014).

Antimicrobial properties of garlic are mainly due to allicin, which is produced on crushing and acts as a broad-spectrum antibacterial agent (Feldberg et al., 1988; Tsao & Yin, 2001).

Preparation of Honey Solution

Honey in its pure form was procured from a local vendor and then subjected to sterilization through membrane filtration with a filter pore size of 0.22 µm. Four different concentrations of honey, that is, 10%, 20%, 30%, and 40% v/v were prepared using distilled water (Molan, 1992; Almasaudi, 2021).

The selection of honey was made owing to the well-known antimicrobial characteristics of honey such as osmosis, acidity, hydrogen peroxide formation, and phytochemicals (Cheesman et al., 201

Preparation of Amoxicillin Stock Solution

A stock solution of amoxicillin powder was prepared at a concentration of 1 mg/mL in sterile distilled water. Serial dilutions were made to provide a series of concentrations (256 µg/mL to 1 µg/mL) for antimicrobial assays (Bush & Bradford, 2016).

Amoxicillin works by interfering with the synthesis of the bacterial cell wall by binding to PBPs, which cause bacterial cell lysis (Munita & Arias, 2016).

Standardization of Bacterial Inoculum

The bacterial suspension was prepared using freshly isolated cultures of E. coli suspended in sterile normal saline solution. The turbidity of the suspension was made equal to that of the 0.5 McFarland standard, which is equivalent to a cell concentration of about 1.5×10^8 CFU/mL (CLSI, 2020).

This suspension was then diluted to create a final inoculum of about 1.5×10^6 CFU/mL (Andrews, 2001).

Antibiotic Susceptibility Testing

The antibiotic susceptibility testing was done through the disk diffusion test according to CLSI (2020) recommendations. Procedure

- A standardized bacterial inoculum was inoculated evenly in the MHA plates.
- The plates were supplemented with amoxicillin and ciprofloxacin disks.
- The MHA plates were then incubated at 37°C for 18–24 hours.
- Zone of inhibition sizes were recorded in millimeters.

The results of testing were interpreted according to the CLSI recommendations. It is widely used because of its simplicity and efficiency (CLSI, 2020).

Determination of Minimum Inhibitory Concentration (MIC)

Antimicrobial sensitivity testing was done using the broth microdilution technique, regarded as the benchmark for determining MIC values (Andrews, 2001).

Procedure:

- Sample solutions were diluted in serial two-fold concentrations in the microtiter plate
- The wells had a combination of M-H broth and the standardized bacterial suspension
- There were controls, including growth and sterility controls
- The plates were incubated at 37 degrees Celsius for 18-24 hours

The MIC value refers to the lowest concentration of the sample solution with no bacterial growth visible.

Checkerboard Assay for Synergy Testing

A test of the interaction effect of garlic extract, honey, and amoxicillin on bacteria using the checkerboard method, an extensively used assay for interactions among antimicrobials, was performed (Odds, 2003; Eliopoulos & Moellering, 2005).

Procedure:

Serial dilutions of two or more drugs were made in the microtiter plates.

Various combinations of the drugs were present in each well.

Bacteria were inoculated into the wells and incubated at 37°C.

The FICI value was determined from:

FICI = [(MIC of drug A in combination / MIC of drug A individually) + (MIC of drug B in combination / MIC of drug B individually)]

Interpretation:

- FICI ≤ 0.5: Synergistic
- 0.5-1: Additive
- 1: Indifferent
- 4: Antagonism

This technique quantitatively determines the interaction of antimicrobial agents and is commonly employed in synergy testing (Odds, 2003).

Result

The antibiotic susceptibility testing (Table 1, Table 2) revealed that **70% of *E. coli* isolates were resistant to amoxicillin**, indicating a high prevalence of resistance. Where as (Table 3 & Table 4), ciprofloxacin showed relatively lower resistance (50%), suggesting that fluoroquinolones may still retain partial effectiveness.

The MIC values obtained in (Table 5) this study indicate that:

- Amoxicillin showed high MIC (**64 µg/mL**), confirming resistance
- Garlic extract exhibited antimicrobial activity (**16 mg/mL**)
- Honey showed inhibitory activity (**10%**)

One of the most important findings of this study is the significant reduction in MIC of amoxicillin (Table 6) when used in combination:

- 8-fold reduction with garlic
- 16-fold reduction with honey
- 32-fold reduction in triple combination

The checkerboard assay (Table 7 & Table 8) confirmed the presence of synergistic interactions:

- Amoxicillin + Garlic → Additive (FICI = 0.5)
- Amoxicillin + Honey → Strong synergy (FICI = 0.375)
- Triple combination → Very strong synergy (FICI = 0.281)

According to standard interpretation (FICI ≤ 0.5), the combinations demonstrated significant synergy, especially the triple combination.

Table 1 Zone of Inhibition of Amoxicillin

Isolate	Amoxicillin	Interpretation
E1	11	Resistant
E2	10	Resistant
E3	12	Resistant
E4	15	Intermediate
E5	9	Resistant
E6	18	Sensitive
E7	11	Resistant
E8	10	Resistant
E9	13	Resistant
E10	17	Sensitive

Table 2 Percentage Distribution of Amoxicillin

Category	Number	Percentage
Resistant	7	70%
Intermediate	1	10%
Sensitive	2	20%

Table 3 Zone of Inhibition of Ciprofloxacin

Isolate	Ciprofloxacin	Interpretation
E1	10	Resistant
E2	12	Resistant

E3	15	Intermediate
E4	11	Resistant
E5	10	Resistant
E6	18	Sensitive
E7	14	Intermediate
E8	11	Resistant
E9	19	Sensitive
E10	17	Sensitive

Table 4 Percentage Distribution of Ciprofloxacin

Category	**Number**	**Percentage**
Resistant	**5**	**50%**
Intermediate	**2**	**20%**
Sensitive	**3**	**30%**

Table 5 Minimum Inhibitory Concentration (MIC) of Individual Agents

Agent	**MIC**
Amoxicillin	64 µg/mL
Garlic	16 mg/mL
Honey	10%

Table 6 MIC Reduction of Amoxicillin in Combination

Treatment	**MIC of Amoxicillin Alone**	**MIC in Combination**	**Fold Reduction**
Amoxicillin + Garlic	64 µg/mL	8 µg/mL	8-fold
Amoxicillin + Honey	64 µg/mL	4 µg/mL	16-fold

Garlic + Honey + Amoxicillin	64 µg/mL	2 µg/mL	32-fold

Table 7 Checkerboard assay

Table 7.1

Treatment	**MIC in combine**	**FIC**
Garlic in combine	4 mg/mL	0.25
Mic of Honey in combine	2.5%	0.25

Table 7.2

Treatment	**MIC of Amoxicillin Alone**	**MIC in Combination**	**FIC**	**FICI**	**Interpretation**
Amoxicillin + Garlic	64 µg/mL	16 µg/mL	**0.25**	**0.5**	Additive
Amoxicillin + Honey	64 µg/mL	8 µg/mL	**0.125**	**0.375**	Strong Synergy

Table 8 Triple Combine

Table 8.1

Agent	**MIC Alone**	**MIC combine**	**FIC**
Amoxicillin	64 µg/mL	2 µg/mL	0.031
Garlic	16 mg/mL	2 mg/mL	0.125
Honey	10%	1.25%	0.125

Table 8.2

Treatment	**FICI**	**Interpretation**
Amoxicillin + Garlic + Honey	0.281	Very Strongly Synergy

Discussion

The findings confirm increasing resistance of *E. coli* to β-lactam antibiotics (Davies & Davies, 2010; Holmes et al., 2016). Garlic extract showed antibacterial activity due to allicin, which inhibits thiol-containing enzymes and disrupts bacterial metabolism (Borlinghaus et al., 2014; Feldberg et al., 1988). Honey demonstrated antimicrobial and antibiofilm activity through multiple mechanisms (Molan, 1992; Kwakman et al., 2010; Lu et al., 2011).

The synergistic interaction observed aligns with previous studies demonstrating enhanced antibiotic efficacy when combined with natural products (Hemaiswarya et al., 2008; Jenkins & Cooper, 2012).

Conclusion

The combination of *Allium sativum* extract, honey, and amoxicillin demonstrated significant synergistic antibacterial activity against MDR *E. coli*. This strategy may improve antibiotic efficacy and reduce resistance.

Further in vivo studies and clinical trials are required.

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Conflict of Interest

The authors declare that there are no conflict of interest related to this work.

References

1. World Health Organization. Global action plan on antimicrobial resistance. WHO; 2015.
2. Kaper JB, Nataro JP, Mobley HL. Pathogenic *Escherichia coli*. Nat Rev Microbiol. 2004;2(2):123–140.

3. Blair JMA, et al. Molecular mechanisms of antibiotic resistance. Nat Rev Microbiol. 2015;13:42–51.
4. Borlinghaus J, et al. Allicin: chemistry and biological properties. Molecules. 2014;19(8):12591–12618.
5. Ankri S, Mirelman D. Antimicrobial properties of allicin from garlic. Microbes Infect. 1999;1(2):125–129.
6. Almasaudi SB. The antibacterial activities of honey. Saudi J Biol Sci. 2021;28(4):2188–2196.
7. Hemaiswarya S, et al. Synergism between natural products and antibiotics. Phytomedicine. 2008;15(8):639–652.
8. Ventola CL. The antibiotic resistance crisis: part 1: causes and threats. *P T.* 2015;40(4):277–283.
9. Prestinaci F, Pezzotti P, Pantosti A. Antimicrobial resistance: a global multifaceted phenomenon. *Pathog Glob Health.* 2015;109(7):309–318.
10. Tadesse DA, et al. Antimicrobial drug resistance in *Escherichia coli* from humans and food animals. *Foodborne Pathog Dis.* 2012;9(6):545–556.
11. Munita JM, Arias CA. Mechanisms of antibiotic resistance. *Microbiol Spectr.* 2016;4(2).
12. Tenover FC. Mechanisms of antimicrobial resistance in bacteria. *Am J Med.* 2006;119(6 Suppl 1):S3–S10.
13. Feldberg RS, et al. In vitro mechanism of inhibition of bacterial growth by allicin. *Antimicrob Agents Chemother.* 1988;32(12):1763–1768.
14. Molan PC. The antibacterial activity of honey. 1. The nature of the antibacterial activity. *Bee World.* 1992;73(1):5–28.
15. Molan PC. The antibacterial activity of honey. 2. Variation in potency. *Bee World.* 1992;73(2):59–76.
16. Kwakman PHS, et al. Medical-grade honey kills antibiotic-resistant bacteria. *FASEB J.* 2010;24(7):2576–2582.
17. Lu J, et al. Manuka-type honeys can eradicate biofilms. *FEMS Immunol Med Microbiol.* 2011;62(3):256–264.
18. CLSI. (2020). *Performance standards for antimicrobial susceptibility testing*.
19. Davies, J., & Davies, D. (2010). Origins of antibiotic resistance. *Microbiology and Molecular Biology Reviews*, 74, 417–433.
20. Holmes, A. H., et al. (2016). Understanding antimicrobial resistance. *The Lancet*, 387, 176–187.
21. Jenkins R, Cooper R. Improving antibiotic activity against wound pathogens with Manuka honey. *PLoS One.* 2012;7(9):e45600.
22. Cheesman MJ, et al. The antimicrobial activity of honey: mechanisms and clinical implications. *Front Microbiol.* 2017;8:569.

23. Yap PSX, et al. Synergistic antibacterial activity between plant extracts and antibiotics. *J Appl Microbiol.* 2014;116(4):1063–1073.
24. Odds FC. Synergy, antagonism, and what the chequerboard puts between them. *J Antimicrob Chemother.* 2003;52(1):1.
25. Eliopoulos GM, Moellering RC. Antimicrobial combinations. In: *Antibiotics in Laboratory Medicine.* 5th ed. 2005.
26. Bush K, Bradford PA. β-lactams and β-lactamase inhibitors. Cold Spring Harb Perspect Med. 2016.
27. Andrews JM. Determination of MIC. J Antimicrob Chemother. 2001.
28. Poirel L, Madec JY, Lupo A, et al. Antimicrobial resistance in *Escherichia coli*. *Microbiol Spectr.* 2018;6(4).
29. Magiorakos AP, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria. Clin Microbiol Infect. 2012;18(3):268–281.
30. Paterson DL, Bonomo RA. Extended-spectrum β-lactamases. Clin Microbiol Rev. 2005;18(4):657–686.
31. Cowan MM. Plant products as antimicrobial agents. Clin Microbiol Rev. 1999;12(4):564–582.
32. Tsao SM, Yin MC. In vitro antimicrobial activity of four diallyl sulphides. *J Med Microbiol.* 2001;50(7):646–649.